Ether knot

Can a nonlinear ether hold a knot of field energy at a fixed size? — register entry H29, step 2

kronspice simulation report · 2026-11-07 · κ2 = κ4 = 1; arrested Newton flow, dt = 0.05 h

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Summary

Hypothesis H29 holds that particles are knots of field energy, held together by their topology. Step 1 (*knotted_light*) found that linear Maxwell keeps a knot's topology but not its energy: the knot links once for ever while flying apart at the speed of light. Step 2, here, asks whether a nonlinear, fluid-like ether can hold a knot by itself, at a size of its own. The ether chosen is a unit-vector field n(x) whose energy is its stiffness plus the magnetic energy of knotted light (Faddeev's model).

- **The knot settles at a fixed size:** extrapolated to zero grid spacing, energy E = 547.5·√(κ2κ4) and stiffness/magnetic ratio E2/E4 = 0.9972 (the virial theorem requires 1), core radius about 1.2 ℓ.
- **Topology intact in 3-D:** two pairs of preimage curves link at 1.0001 and 1.0002 — Hopf charge 1.
- **Controls behave as Derrick's theorem predicts:** the magnetic term alone spreads the knot (rms radius 1.003 → 5.125 ℓ, energy 311.713 → 0.087); the stiffness alone shrinks it until it unties through the grid.
- **The knot has a tone:** struck, it rings by squashing at ω ≈ 1.36–1.42 c/ℓ with Q ≈ 6–7; breathing is broad (ω ≈ 0.43–0.50 c/ℓ, Q ≈ 2) and twisting hardly rings (Q ≈ 1–2).
- **Gap to the literature recorded:** Battye & Sutcliffe's 100³ lattice gave E = 504 (spacing not stated), 8 % below this run.
- **Not fixed by the model:** κ2 and κ4; no measured ether stiffness sets them, so size, energy and frequency are ratios, not yet numbers.

What is being tested

Hypothesis (H29). Particles are knots of field energy, held together by their topology (a knot cannot untie without being cut); Kelvin's vortex atoms (*Phil. Mag.* 34, 15, 1867) in field form. Step 2 asks whether a nonlinear ether holds a knot at rest without any inflow of energy.

Standard prediction. Derrick (1964): no static lump of a linear field exists in three dimensions. Under a change of size x → λx the stiffness term grows as λ and the magnetic term falls as 1/λ, so alone the magnetic term spreads the knot and the stiffness shrinks it; together they fix a size where the two are equal. A holding knot needs a nonlinear term (Skyrme 1961; Faddeev & Niemi, *Nature* 387, 58, 1997).

Which ether? (literature survey from the README)

candidate	what the literature says	holds a knot at rest?
Light made nonlinear: Born–Infeld (1934), Euler–Heisenberg (1936, seen as light-by-light scattering by ATLAS, Nature Phys. 13, 852, 2017)	the only finite-energy, source-free, stationary Maxwell–Born–Infeld solution is zero (Kiessling, arXiv:1107.2333, Theorem 1)	no
an ordinary superfluid (Gross–Pitaevskii), Kelvin's vortex atoms	all 1,458 simulated vortex knots untie by reconnection at the vortex cores, where the field vanishes (Kleckner, Kauffman & Irvine, Nature Phys. 12, 650, 2016)	no
a fluid whose order parameter never vanishes: a unit vector n(x); a charged two-component superfluid reduces to it (Babaev, Faddeev & Niemi, PRB 65, 100512, 2002)	Faddeev's knots; stable, computed by Battye & Sutcliffe (PRL 81, 4798, 1998): Q = 1 energy 504 on their 100³ lattice	yes: tested here

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The model

The ether's energy is

$$E = \kappa_2 \int (\partial_i n)^2 \, d^3x \;+\; \kappa_4 \int |B_n|^2 \, d^3x, \qquad B_{n,k} = \tfrac{1}{2}\, \varepsilon_{kij}\, n \cdot (\partial_i n \times \partial_j n)$$

For Rañada's Hopf map, the starting field, B_n is exactly Rañada's knotted magnetic field times −16 at every point (tested). So the second term is the knotted light's magnetic energy, the term step 1 already had, and the first term (E2) is the new ingredient: the ether's stiffness. Units: κ2 = κ4 = 1, so lengths are in ℓ = √(κ4/κ2) and energies in √(κ2κ4).

The Q = 1 knot is axially symmetric, with the field turning once round the axis: n = $R_z(\varphi)$ m(ρ, z). The energy is therefore minimized on the (ρ, z) half-plane, about a thousand times cheaper than in 3-D, with m kept a unit vector. Relaxation is arrested Newton flow (as Battye & Sutcliffe): the field accelerates downhill and is stopped whenever the energy rises. The start is Rañada's map pulled in smoothly to reach the vacuum inside the box. The relaxed knot is then rebuilt on a 3-D grid to check its energy and topology.

Results

1. Stiffness + magnetic: grid and box study

box (ℓ)	h (ℓ)	steps	E	E2 (stiffness)	E4 (magnetic)	E2/E4	core radius (ℓ)	rms radius (ℓ)
10	0.1	4000	541.084	261.147	279.937	0.9329	1.150	1.816
16	0.1	6000	540.244	262.109	278.135	0.9424	1.150	1.870
16	0.07	8000	544.004	268.062	275.942	0.9714	1.225	1.911
16	0.05	12000	545.701	270.656	275.046	0.9840	1.225	1.929
h → 0	**(h = 0.07, 0.05)**		**547.5**			**0.9972**		

- The box edge hardly matters (about 1 part in 500 from 10 ℓ to 16 ℓ); the grid does. As h → 0 the stiffness and magnetic terms become equal to 0.3 % — Derrick's balance, reached by the relaxation without being imposed.
- Battye & Sutcliffe's 100³ lattice gave 504 (their spacing is not stated). Coarse grids give lower energies here too (540 at h = 0.1), so the 8 % gap is recorded, not explained away; a continuum value from the literature in the same normalization is still to be checked.

2. Controls (box 10 ℓ, h 0.1 ℓ, from the same start)

magnetic only (κ2 = 0)	step 0	1500	3000	4500	6000
E	311.713	6.835	0.600	0.148	0.087
rms radius (ℓ)	1.003	3.229	4.182	5.124	5.125

stiffness only (κ4 = 0)	step 0	100	200	300	500	700	1000
E	270.795	236.362	129.480	66.072	37.233	14.569	4.943
core radius (ℓ)	0.950	0.750	0.350	2.650	2.550	5.650	1.750
m_z at core	−0.997	−0.994	−0.016	+0.754	+0.849	+0.979	+0.986

Without the ether (κ2 = 0, knotted light alone) the knot spreads: step 1's result, now as an energy minimization. With stiffness alone it shrinks; by step 200 the core reaches the grid spacing, m_z at the core leaves −1 and the knot unties through the lattice. In the continuum it would shrink to a point; a grid cannot hold topology below its own spacing.

3. The relaxed knot in 3-D

The finest knot (box 16 ℓ, h = 0.05) was rebuilt on 101³ points over ±8 ℓ. Two pairs of preimage curves of n (the field lines of B_n), each closed at its nearest return, link at **1.0001** (gap 0.019 ℓ) and **1.0002** (gap 0.011 ℓ): Hopf charge 1, the knot intact. The 3-D energy reads E2 297.28 + E4 255.40 = 552.68 against 270.656 + 275.046 in the axisymmetric solve: the ±8 ℓ box cuts the tail of a knot relaxed in a 16 ℓ box (stiffness high) and the 3-D grid (h = 0.16) is about three times coarser (magnetic term low by O(h²)). A knot relaxed in an 8 ℓ box and rebuilt the same way matched to 0.5 % in stiffness (258.7 against 260.0), with the magnetic term 6 % low (README).

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4. Vibrations: the knot's own tone

Small motions x about the relaxed knot obey $2K\,\ddot{x} = -H\,x$, from Faddeev's Lagrangian (kinetic energy $\int|\dot{n}|^2 + \Sigma|\dot{n} \times \partial_i n|^2$): H is the energy's Hessian on the sphere and K the kinetic form, both built for motions that keep the axial symmetry. Two motions cost nothing and come out at $\omega^2 \approx 0$ (sliding along z, turning the field about $\hat{z}$): 2.2×10^{-4} and 9.9×10^{-8} (box 16 ℓ, h 0.15), 1.6×10^{-4} and -3.5×10^{-7} (box 24 ℓ), 1.5×10^{-4} and 1.6×10^{-7} (box 16 ℓ, h 0.1), against the real modes' ~1.

Small waves of this ether travel at c with no lowest frequency, so a vibrating knot leaks energy and has no sealed tone. In a box the low eigenmodes are standing waves of the box: the lowest falls 0.428 → 0.324 → 0.252 c/ℓ as the box grows 8 → 12 → 16 ℓ, and its share within 3 ℓ of the knot falls 42 % → 9 % → 2 % (README; vib_output.txt gives 0.2519 c/ℓ and 0.02 for box 16 ℓ). So the knot was struck instead, with three localized pushes (breathing, squashing, twisting, under $\exp(-r^2/8\ell^2)$, zero modes removed), left to ring, and its motion fitted to damped cosines (matrix pencil) up to the return of the first echo from the wall. Strongest component of each push:

push	box 16 ℓ, h 0.15 ℓ	box 24 ℓ, h 0.15 ℓ	box 16 ℓ, h 0.10 ℓ
squashing	**ω 1.416, Q 7.0**	**ω 1.415, Q 7.0**	**ω 1.385, Q 6.0**
breathing	ω 0.429, Q 1.9	ω 0.447, Q 1.9	ω 0.498, Q 2.0
twisting	ω 0.931, Q 1.7	ω 0.859, Q 1.1	ω 0.829, Q 1.3

ω in c/ℓ. Records to t = 20 ℓ/c (box 16 ℓ) and 36 ℓ/c (box 24 ℓ). Weaker components are listed in vib_output.txt.

- **Squashing is the knot's tone.** A bigger box leaves it unchanged (1.416 → 1.415); the finer grid lowers it 2 %; extrapolated in h^2, about 1.36 c/ℓ. With Q ≈ 6–7 the energy falls by e in about one cycle.
- **Breathing**, the size change Derrick's balance fixes, is a broad resonance near 0.45–0.5 c/ℓ, gone in about two cycles (Q ≈ 2). **Twisting** hardly rings: no tone of its own.
- The frequency is set by the knot's size: $f \approx 1.4\ c/(2\pi\ell)$, $\omega\cdot(\text{core radius})/c \approx 1.7$. For ℓ = 1 m it would be 67 MHz, for ℓ = 1 fm, 6.7×10^{22} Hz. The squashing-to-breathing ratio, about 3, needs no constant at all.

Validation

test	checks	held to
TestEtherCurvatureIsKnottedLight	B_n of Rañada's map = −16 × his knotted B, at four points	10^{-6}
TestGradient	analytic gradients of the 3-D and axisymmetric energies against finite differences	10^{-6}; 10^{-5}
TestAxiEqualsThreeD	axisymmetric and 3-D energies of the same field (E4 converges as h^2)	1 %; 6 %
TestVibrations	coarse grid: two zero modes at $\omega^2 \approx 0$; a squashing push rings at ω ≈ 1.45 c/ℓ (zero crossings)	0.02; 0.15
TestEtherHoldsKnot	coarse grid: both terms near the virial balance with a core ~ℓ; magnetic only spreads; stiffness only shrinks, then unties	0.2; ×1.1; ×0.9, 5 %

Limits of the model

- **κ2 and κ4 are not fixed.** The knot's size is $\ell = \sqrt{(\kappa4/\kappa2)}$ and its energy $547.5\cdot\sqrt{(\kappa2\kappa4)}$; no measured “ether stiffness” exists to set them. Matching one particle's mass and size uses both numbers, so the first prediction would be a ratio between two particles (e.g. the Q = 2 knot against Q = 1: Battye & Sutcliffe 835 against 504).
- **It is not electromagnetism.** n is an ordinary field with its own energy. B_n has the form of a magnetic field, but how n would couple to electric charges, and whether such a knot carries charge or spin ½, is open.
- Only axially symmetric relaxation and vibrations are computed; the 8 % gap to Battye & Sutcliffe is not resolved.

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Status

H29 step 2 is **reproduced**: a nonlinear ether whose order parameter never vanishes holds the knot at a fixed size and energy, with its linking number intact and no inflow needed. That is the standard result of Faddeev & Niemi (1997), now reproduced in kronspice and tied to the knotted light of step 1. Next steps: the Q = 2 knot (a prediction with no new constant), vibrations that break the axial symmetry, and a Faddeev ether coupled to the field solver *em/fdtd.go*.

Reproducing the results

```
go test -short ./examples/ether_knot     # field link, gradients, 2-D vs 3-D (< 1 s)
go test ./examples/ether_knot            # + the relaxation and controls (~15 s)
go run ./examples/ether_knot             # grid study, controls, 3-D check (~30 min on 2 cores)
go run ./examples/ether_knot -ring examples/ether_knot/vib     # vibrations, ~8 min on 2 cores
python3 examples/ether_knot/ring.py examples/ether_knot/vib/ring_*.json
python3 examples/ether_knot/modes.py examples/ether_knot/vib/ops_b16_h0.15.txt
python examples/ether_knot/viz/make_report.py examples/ether_knot/ether_knot_report.pdf
```

Sources: Faddeev & Niemi, Nature 387, 58 (1997); Battye & Sutcliffe, PRL 81, 4798 (1998); Babaev, Faddeev & Niemi, PRB 65, 100512 (2002); Kleckner, Kauffman & Irvine, Nature Phys. 12, 650 (2016); Kiessling, arXiv:1107.2333; ATLAS, Nature Phys. 13, 852 (2017); Derrick, J. Math. Phys. 5, 1252 (1964); Rañada, Lett. Math. Phys. 18, 97 (1989); Hua & Sarkar, IEEE Trans. ASSP 38, 814 (1990); Volovik, The Universe in a Helium Droplet (2003).

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